Truck Turnaround Time (TAT) Analysis

Key Findings and Optimisation Model Results

FMCG warehouse operations | 4 warehouses | 2,500 truck visits | April to September 2026 | Target TAT: 3.0 hours (assumed)

4.32 h	21.1%	84 min	79.3 min
Average TAT (median 4.02 h)	Trucks within 3.0 h target	Average dock wait (32% of TAT)	Cut needed per truck to reach 3.0 h

A. Key Findings

1. Network performance is far from the target

- Average TAT is **4.32 hours (259 minutes)**, median 4.02 hours. Only **527 of 2,500 trucks (21.08%)** finish within 3.0 hours; **1,973 trucks (78.92%)** breach it.
- The distribution is right-skewed (skewness 1.29) with a tail up to 13.30 hours. 65.0% of trucks spend 3 to 6 hours inside the warehouse and 13.9% spend more than 6 hours.
- The IQR rule (upper limit 8.07 hours) flags 82 trips (3.28%). All 82 were logged as Dock Congestion and waited 371 minutes on average at the dock.
- No month meets the target. Monthly average TAT ranges from 3.89 hours (August) to 4.72 hours (April), and follows the volume handled on the sampled days (about 72 trucks per day in April against about 48 in August).

2. Dock waiting and loading consume three quarters of the visit

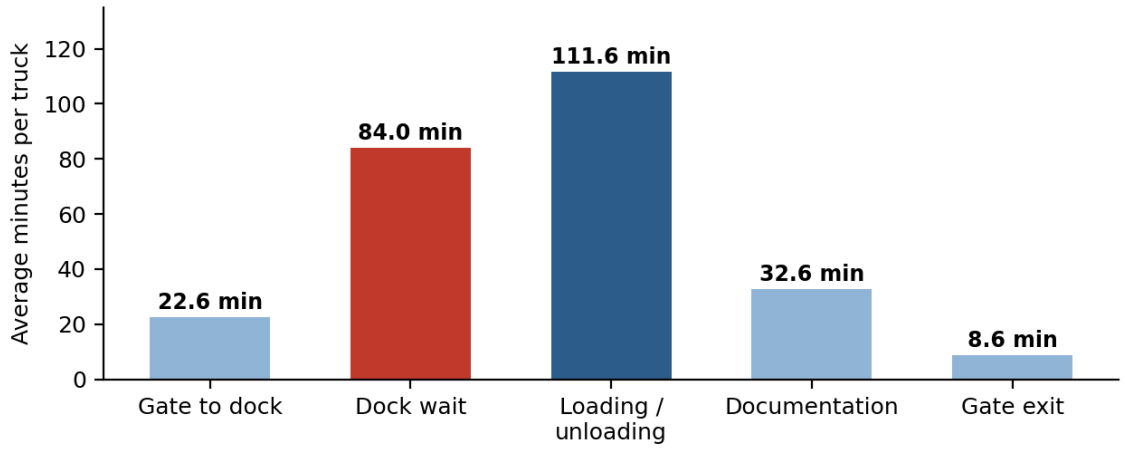


Figure 1. Average time per truck in each stage

- Loading/unloading (111.6 min, 43%) and dock waiting (84.0 min, 32%) together take 195.5 minutes, more than the whole 3.0-hour target.
- Dock wait is the main non-value-added stage: 49.4% of trucks wait more than 60 minutes, and dock wait has a 0.88 correlation with total TAT.
- Gate-to-dock (9%), documentation (13%) and gate exit (3%) are small, so gate and paperwork improvements alone cannot close the gap.

3. Dock congestion is the dominant root cause

- **1,531 visits (61.2%)** carry a logged delay. Of these, **Dock Congestion is 1,095 trips (71.5%)**.
- Dock Congestion also costs the most: about **2,573 extra truck-hours**. Dock-wait related reasons together account for about 2,707 of roughly 3,053 extra truck-hours (88.7%).
- Trips with no recorded delay average 3.10 hours, close to the target, so the standard process has little slack.

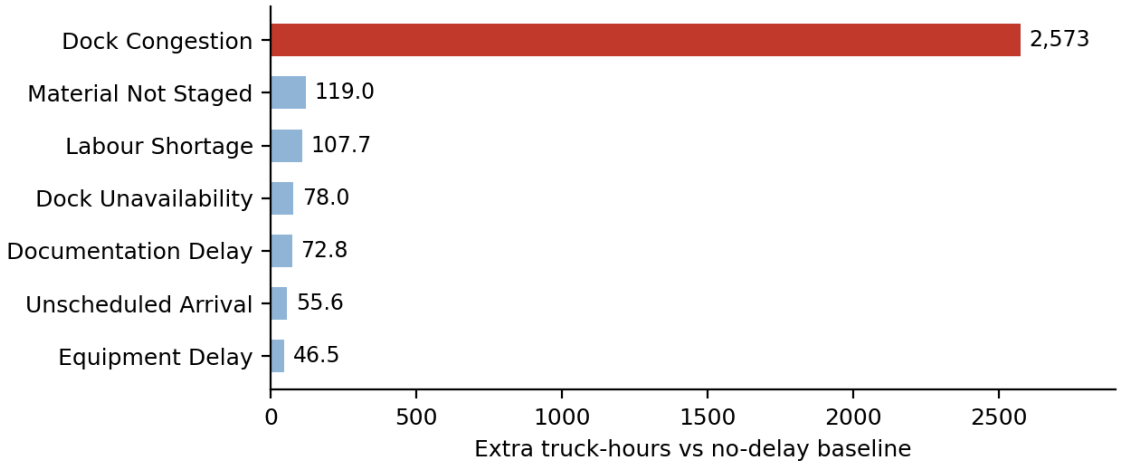


Figure 2. Extra truck-hours by primary delay reason (against the no-delay baseline)

4. Warehouse performance follows workload per dock

Warehouse	Docks	Trucks	Trucks per dock per day	Dock utilisation*	Avg dock wait (min)	Avg TAT (hrs)
Patna Distribution	3	592	4.70	74.3%	117.4	4.85
Kolkata Regional	5	842	4.01	56.2%	88.0	4.43
Jamshedpur Central	4	627	3.73	45.3%	63.5	3.99
Ranchi Distribution	3	439	3.48	41.1%	60.4	3.88

*Calculated on a 24-hour dock day. Trucks arrive only between 08:00 and 19:59, so utilisation during working hours is higher than shown.

- TAT and dock wait rise in the same order as workload per dock across all four sites. Patna (3 docks, highest load per dock) is the first candidate for extra capacity or slot control.

5. Arrival pattern: the queue lags the arrival peaks

- **59.0% of trucks (1,475)** arrive in four peak hours (10:00 to 11:59 and 14:00 to 15:59): 852 in the morning peak and 623 in the afternoon peak.
- Dock wait is similar for peak (85.0 min) and off-peak (82.5 min) arrivals because the queue builds through the day: about **28 minutes** for 08:00 to 09:59 arrivals, 95 to 110 minutes from midday to 15:59, and a maximum of **135 minutes** for 17:00 arrivals.

6. Transporter type does not explain delay

- Spot carriers average 4.40 hours against 4.29 hours for contract carriers, a gap of about 6 minutes that is **not statistically significant** (Welch t-test, p = 0.12). Dock queues affect all carriers alike.

B. Optimisation Model Results

The optimisation was built in two layers. Layer 1 tests how much dock allocation alone can recover. Layer 2 works out how many minutes each stage must be cut to reach 3.0 hours at the lowest cost.

Layer 1: Dock allocation simulation

Step 1, diagnosis. For each dock, a truck is flagged as arriving at a busy dock if the previous truck at that dock had not yet finished loading.

- **1,482 trucks (59.3%)** reached a busy dock and waited **125.1 minutes** on average (median 104). The other 1,018 trucks (40.7%) found the dock free and waited only **24.0 minutes** (median 15).
- The gap stays large (48.8 vs 21.9 minutes) after excluding dock-related delay reasons, so the queue behind an occupied dock is a genuine driver of waiting.

Step 2, simulation. Within each warehouse and day, trucks are served first come, first served, and each truck goes to the dock that becomes free first. Loading times, other stage times and arrival times are unchanged; only dock wait is recalculated.

Metric	Actual	Simulated (first free dock)	Change
Average dock wait (min)	83.95	45.76	-38.2 min
Average TAT (hours)	4.32	3.69	-0.63 hours
Trucks within 3.0 hours	21.08%	36.12%	+15.0 points

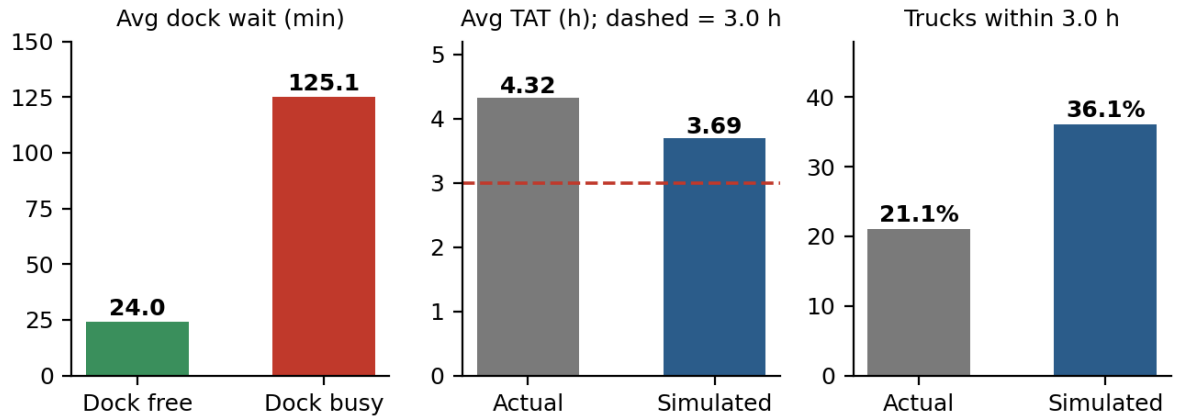


Figure 3. Layer 1: busy vs free dock wait, and actual vs simulated performance

- **Result:** better dock allocation alone recovers about 38 minutes per truck, but the average TAT of 3.69 hours is still above 3.0 hours. A queue remains even with ideal allocation, so extra dock capacity, slot booking and faster bay readiness are also needed.

Layer 2: Stage-wise cost minimisation (linear programming)

Formulation. Choose the minutes to cut from each of four stages so that total cost is minimised, subject to (a) total cut at least the required cut, and (b) no stage cut beyond its best achievable level, which is assumed to be the network's own 25th percentile. Solved with linear programming (SciPy linprog, HiGHS).

Stage	Current avg (min)	Best level, 25th pct (min)	Max possible cut (min)
Gate to dock	22.6	15.0	7.6
Dock wait	84.0	17.0	67.0
Loading	111.6	82.0	29.6
Documentation	32.6	25.0	7.6
Total (incl. gate exit 8.6 min, not cut)	259.3 (4.32 h)	Target 180 (3.0 h)	111.7 max; **79.3 required**

The target needs about 71% of the maximum possible saving. Dock wait has the largest room for improvement: 67 minutes, 60% of the maximum.

Cost scenarios and optimal plans

Relative cost units per minute cut (assumed, for comparison between scenarios only): A Base case [gate 3, dock wait 1, loading 4, documentation 2]; B Loading is cheap [3, 1, 1.5, 2]; C Dock wait is hard [3, 5, 4, 2].

Scenario	Gate (min cut)	Dock wait (min cut)	Loading (min cut)	Documentation (min cut)	Total cost (units)
A: Base case	4.8	67.0	0.0	7.6	96.5
B: Loading is cheap	0.0	67.0	12.4	0.0	**85.5**
C: Dock wait is hard	7.6	34.5	29.6	7.6	329.1

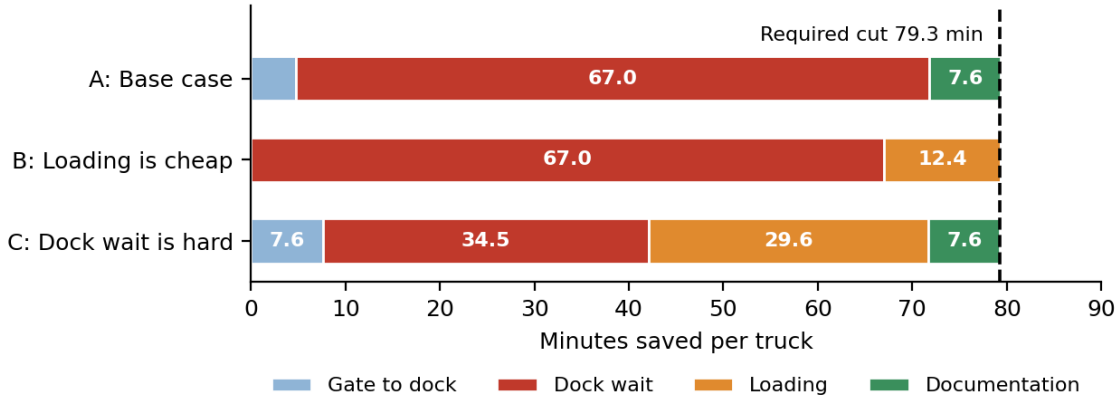


Figure 4. Cheapest plan to reach the 3.0-hour target, by cost scenario

- **Scenarios A and B:** the cheapest plan always cuts dock wait by its full 67 minutes, then uses gate, documentation or loading only for the remaining 12 minutes.
- **Scenario C:** if dock wait is hard to reduce, it contributes only 34.5 minutes, every other stage must be cut to its limit, and the cost rises to 329.1 units, about **3.4 times** the base-case cost.
- **Conclusion:** dock wait is the stage to attack first. Reducing it is the cheapest way to close the 79-minute gap, and the plan becomes much more expensive and less realistic if it is not reduced.

What the two layers say together

- Layer 2 asks for a dock-wait cut of about 67 minutes (to roughly 17 minutes). Layer 1 shows that first-free-dock allocation delivers about 38 minutes (to roughly 46 minutes). The remaining gap should come from dock capacity, appointment/slot booking to flatten the morning surge, and faster bay readiness.
- Priority site: **Patna** (3 docks, 4.7 trucks per dock per day, 117 minutes average dock wait).

Limitations

The data is synthetic, generated with a yard-queue simulation (42 sampled operating days, about 60 trucks per day), so the relationships found must be validated on real warehouse data. The 3.0-hour target is an assumed project benchmark, and the optimisation costs are assumed relative units, not currency. Best achievable stage levels are taken from the network's own 25th percentile.